

SHETH L.U.J AND SIR M.V. COLLEGE
SUBJECT NAME: Applied Business Analytics Practical
Practical 3. Finding Relationships Among Variables:

Aim: Finding Relationships Among Variables:

- a. Use a dataset with multiple variables and perform correlation analysis to Determine the strength and direction of relationships between pairs of variables.
- b. Apply regression analysis to identify the relationship between an independent variable (e.g., advertising expenditure) and a dependent variable (e.g., sales revenue).

Function/Method Description

Function / Method	Description
pd.read_csv()	
files.upload()	Allows the user to upload the dataset directly into Google Colab.
next(iter(uploaded))	Gets the name of the uploaded file automatically.
df.head()	Displays the first five records of the dataset for initial inspection.
df.info()	Displays information about the dataset, including columns, data types, and number of non-null values.
df.corr()	Calculates the correlation coefficients between numerical variables.
plt.figure()	Creates a figure with the specified size for visualization.
sns.heatmap()	Creates a heatmap to visually represent the correlation matrix.
LinearRegression()	Creates a linear regression model to study the relationship between advertising expenditure and sales.
model.fit()	Trains the regression model using the independent variable and dependent variable.
model.predict()	Predicts sales values using the trained regression model.
r2_score()	Calculates the R-squared value, which

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	measures how well the regression model explains the variation in sales.
mean_squared_error()	Calculates the mean squared error between actual and predicted sales values.
plt.scatter()	Creates a scatter plot showing the actual relationship between TV advertising budget and sales.
plt.plot()	Draws the regression line representing the predicted relationship.
plt.xlabel()	Sets the label for the X-axis.
plt.ylabel()	Sets the label for the Y-axis.
plt.title()	Adds a title to the graph.
plt.legend()	Displays labels for the actual data and regression line.
plt.grid()	Adds grid lines to make the graph easier to interpret.
plt.show()	Displays the generated graph.
round()	Rounds numerical results to a specified number of decimal places.

Code:

```
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
from google.colab import files
from sklearn.linear_model import LinearRegression
from sklearn.metrics import mean_squared_error, r2_score
```

```
uploaded = files.upload()
```

```
filename = next(iter(uploaded))
```

```
df = pd.read_csv(filename)
```

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```
print("DATASET LOADED SUCCESSFULLY")
print("=" * 60)

print("\nFirst 5 Records:")
print(df.head())

print("\nDataset Information:")
print(df.info())

correlation = df.corr(numeric_only=True)

print("\nCorrelation Matrix:")
print(correlation)

plt.figure(figsize=(8, 6))
sns.heatmap(
    correlation,
    annot=True,
    cmap="coolwarm",
    fmt=".2f"
)

plt.title("Correlation Matrix")
plt.tight_layout()
plt.show()

X = df[["TV Ad Budget ($)"]]
y = df["Sales ($)"]

model = LinearRegression()

model.fit(X, y)

y_pred = model.predict(X)

print("\nLINEAR REGRESSION RESULTS")
print("=" * 60)
```

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```
print("Intercept:", round(model.intercept_, 4))
print("Coefficient:", round(model.coef_[0], 4))

print("\nRegression Equation:")
print(
    f"Sales = {model.intercept_:.2f} + "
    f"{model.coef_[0]:.4f} × TV Ad Budget"
)

r2 = r2_score(y, y_pred)

mse = mean_squared_error(y, y_pred)

print("\nR-squared:", round(r2, 4))
print("Mean Squared Error:", round(mse, 4))

plt.figure(figsize=(8, 6))

plt.scatter(
    X,
    y,
    label="Actual Sales"
)

plt.plot(
    X,
    y_pred,
    label="Regression Line"
)

plt.xlabel("TV Advertising Budget ($)")
plt.ylabel("Sales ($)")
plt.title("TV Advertising Budget vs Sales")

plt.legend()
plt.grid(True)
plt.tight_layout()
```

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```
plt.show()

print("\nCONCLUSION")
print("=" * 60)

print(
    f"The correlation between TV advertising budget and sales is "
    f"{correlation.loc['TV Ad Budget ($)', 'Sales ($)']:.2f}."
)

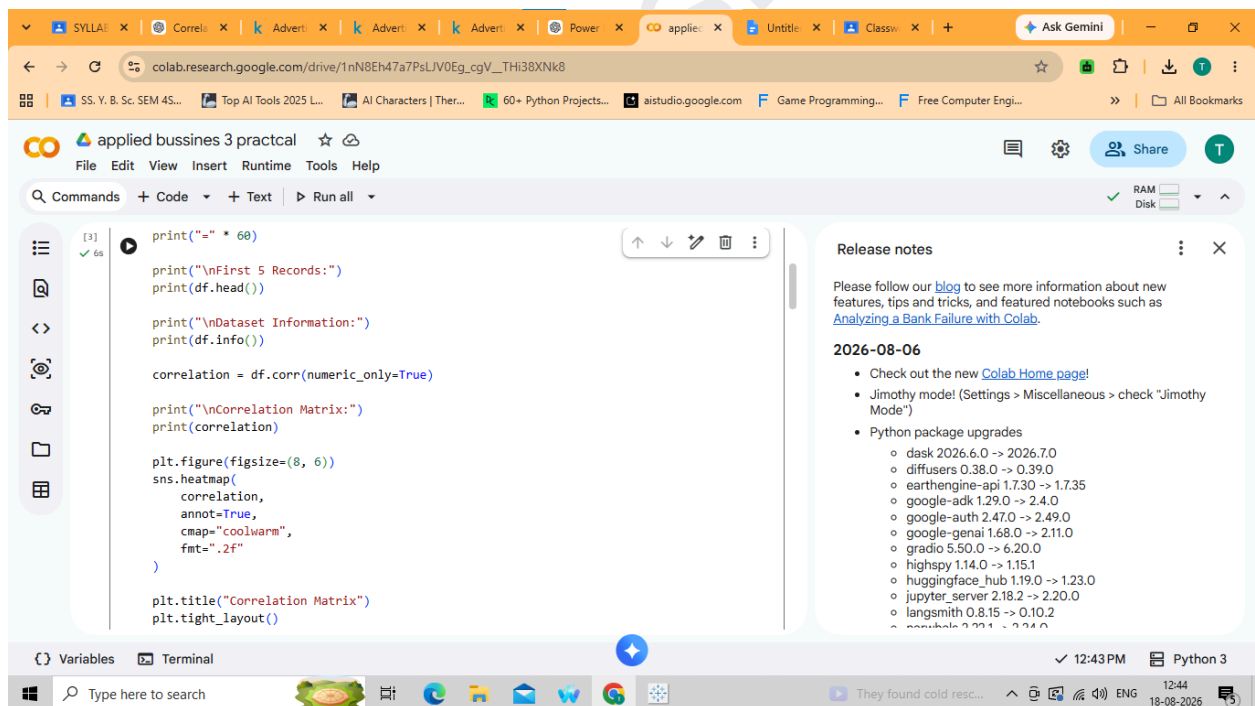
print(
    f"The R-squared value is {r2:.2f}, indicating that approximately "
    f"{r2 * 100:.2f}% of the variation in sales is explained by "
    f"TV advertising expenditure."
)

print(
    "Therefore, TV advertising expenditure has a positive relationship "
    "with sales."
)
```

Screenshots Code:

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The image displays two screenshots of a Google Colab notebook titled "applied bussines 3 practcal". The notebook is running Python code for a linear regression model.

Top Screenshot: The code defines a linear regression model, fits it to the data, and prints the intercept and coefficient. The output shows the intercept as 2.2 and the coefficient as 0.4.

```
[3] plt.title("Correlation Matrix")
plt.tight_layout()
plt.show()

X = df[["TV Ad Budget ($)"]]
y = df["Sales ($)"]

model = LinearRegression()

model.fit(X, y)

y_pred = model.predict(X)

print("\nLINEAR REGRESSION RESULTS")
print("=" * 60)

print("Intercept:", round(model.intercept_, 4))
print("Coefficient:", round(model.coef_[0], 4))

print("\nRegression Equation:")
print(f"Sales = {model.intercept_:.2f} + "
      f"{model.coef_[0]:.4f} x TV Ad Budget")
```

Bottom Screenshot: The code calculates the R-squared value and the Mean Squared Error (MSE), and then plots the scatter of actual sales versus predicted sales.

```
[3] print("Intercept:", round(model.intercept_, 4))
print("Coefficient:", round(model.coef_[0], 4))

print("\nRegression Equation:")
print(f"Sales = {model.intercept_:.2f} + "
      f"{model.coef_[0]:.4f} x TV Ad Budget")

r2 = r2_score(y, y_pred)

mse = mean_squared_error(y, y_pred)

print("\nR-squared:", round(r2, 4))
print("Mean Squared Error:", round(mse, 4))

plt.figure(figsize=(8, 6))

plt.scatter(
    X,
    y,
    label="Actual Sales"
)
```

Both screenshots show a "Release notes" panel on the right side of the notebook interface, which includes information about new features, tips, and tricks, and a list of Python package upgrades.

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The screenshot shows a Google Colab notebook titled "applied bussines 3 practical". The code in the cell is as follows:

```
[3] label="Actual Sales"

plt.plot(
    X,
    y_pred,
    label="Regression Line"
)

plt.xlabel("TV Advertising Budget ($)")
plt.ylabel("Sales ($)")
plt.title("TV Advertising Budget vs Sales")

plt.legend()
plt.grid(True)
plt.tight_layout()
plt.show()

print("\nCONCLUSION")
print(" " * 60)

print(
    f"The correlation between TV advertising budget and sales is "
```

The output shows a scatter plot with a regression line. The x-axis is labeled "TV Advertising Budget (\$)" and the y-axis is labeled "Sales (\$)". The plot title is "TV Advertising Budget vs Sales". The legend indicates the regression line. The conclusion section is currently blank.

The screenshot shows the same Google Colab notebook, but the code is more complete. The output shows a scatter plot with a regression line. The x-axis is labeled "TV Advertising Budget (\$)" and the y-axis is labeled "Sales (\$)". The plot title is "TV Advertising Budget vs Sales". The legend indicates the regression line. The conclusion section is filled with text:

```
print(
    f"The correlation between TV advertising budget and sales is "
    f"{correlation.loc['TV Ad Budget ($)', 'Sales ($)']:.2f}."
)

print(
    f"The R-squared value is {r2:.2f}, indicating that approximately "
    f"{r2 * 100:.2f}% of the variation in sales is explained by "
    f"TV advertising expenditure."
)

print(
    "Therefore, TV advertising expenditure has a positive relationship "
    "with sales."
)
```

The output shows a scatter plot with a regression line. The x-axis is labeled "TV Advertising Budget (\$)" and the y-axis is labeled "Sales (\$)". The plot title is "TV Advertising Budget vs Sales". The legend indicates the regression line. The conclusion section is filled with text.

Outputs :

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```
Choose Files Advertising ... d Sales.csv
Advertising Budget and Sales.csv(text/csv) - 4802 bytes, last modified: 8/18/2026 - 100% done
Saving Advertising Budget and Sales.csv to Advertising Bu
DATASET LOADED SUCCESSFULLY
colab.research.google.com - To exit full screen, press Esc

First 5 Records:
  Unnamed: 0  TV Ad Budget ($)  Radio Ad Budget ($)  Newspaper Ad Budget ($)  \
0           1           230.1           37.8           69.2
1           2           44.5           39.3           45.1
2           3           17.2           45.9           69.3
3           4           151.5           41.3           58.5
4           5           180.8           10.8           58.4

Sales ($)
0           22.1
1           10.4
2            9.3
3           18.5
4           12.9

Dataset Information:
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 200 entries, 0 to 199
Data columns (total 5 columns):
#   Column              Non-Null Count  Dtype
---  ---
0   Unnamed: 0           200 non-null   int64
1   TV Ad Budget ($)     200 non-null   float64
2   Radio Ad Budget ($)  200 non-null   float64
3   Newspaper Ad Budget ($) 200 non-null   float64
4   Sales ($)            200 non-null   float64
dtypes: float64(4), int64(1)
memory usage: 7.9 KB
None

Correlation Matrix:
  Unnamed: 0  TV Ad Budget ($)  Radio Ad Budget ($)  \
Unnamed: 0    1.000000      0.017715      -0.110680
TV Ad Budget ($)  0.017715      1.000000      0.054809
Radio Ad Budget ($) -0.110680      0.054809      1.000000
Newspaper Ad Budget ($) -0.154944      0.056648      0.354104
Sales ($)      -0.051616      0.782224      0.576223
```

The screenshot shows a Google Colab notebook with the same code and output as the first block. At the bottom, a heatmap titled "Correlation Matrix" is displayed, showing the relationships between the variables. The diagonal elements are all 1.0, indicating perfect self-correlation. The off-diagonal elements show the correlation between different variables: TV Ad Budget (\$) and Radio Ad Budget (\$) have a correlation of approximately 0.0177, TV Ad Budget (\$) and Newspaper Ad Budget (\$) have a correlation of approximately -0.1107, TV Ad Budget (\$) and Sales (\$) have a correlation of approximately -0.0516, Radio Ad Budget (\$) and Newspaper Ad Budget (\$) have a correlation of approximately 0.0548, Radio Ad Budget (\$) and Sales (\$) have a correlation of approximately 0.7822, and Newspaper Ad Budget (\$) and Sales (\$) have a correlation of approximately 0.3541. The heatmap uses a color scale from blue (negative correlation) to red (positive correlation).

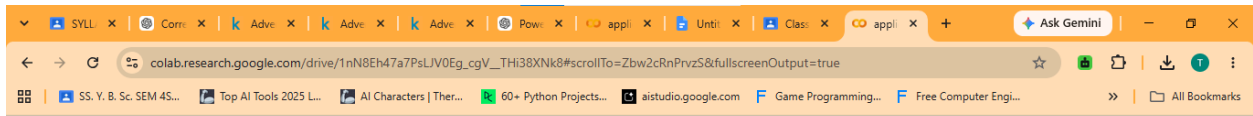
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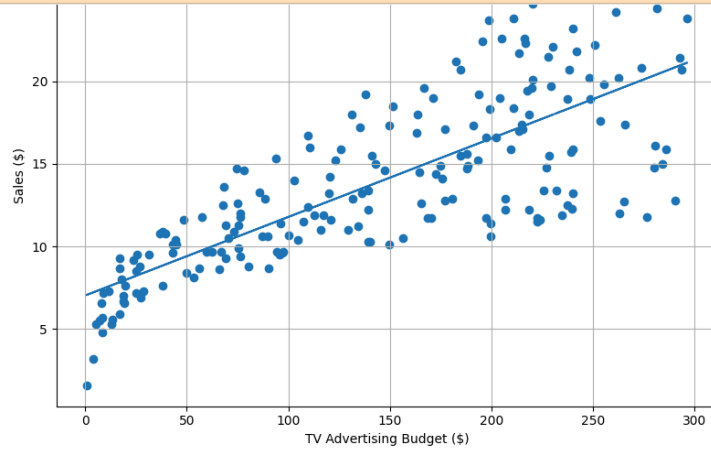
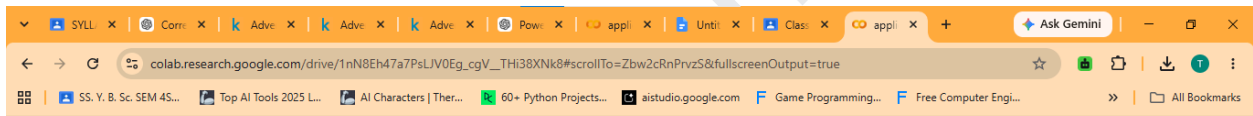
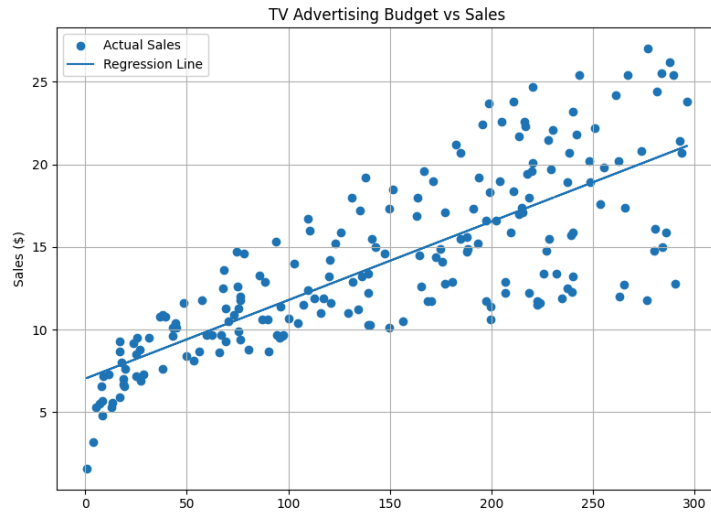


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R-squared: 0.6119
Mean Squared Error: 10.5127



CONCLUSION

=====

The correlation between TV advertising budget and sales is 0.78.
The R-squared value is 0.61, indicating that approximately 61.19% of the variation in sales is explained by TV advertising expenditure.
Therefore, TV advertising expenditure has a positive relationship with sales.



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